

Data Visualization

Data visualization is a crucial component of Exploratory Data Analysis (EDA) as it helps transform raw data into meaningful graphical representations. Visualizations make it easier to identify patterns, trends, relationships, distributions, and anomalies within the dataset.

In this project, various visualization techniques were used to analyze the relationship between advertising expenditure and product sales.

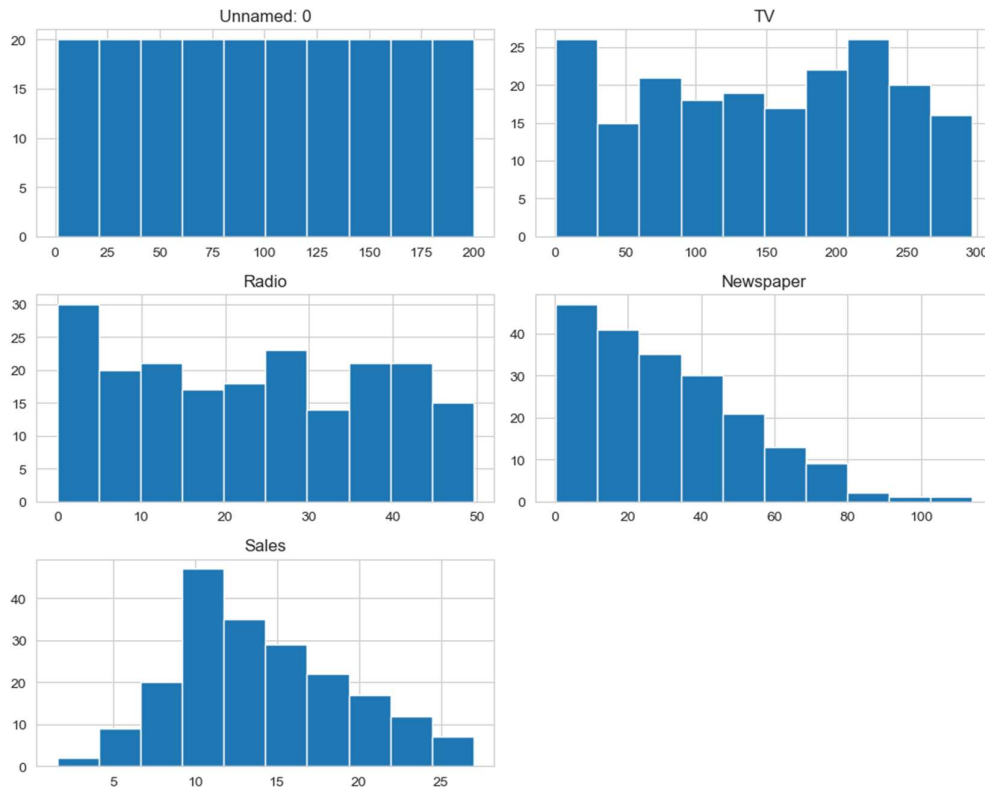
1. Histogram

Histograms were used to examine the distribution of numerical variables such as TV, Radio, Newspaper, and Sales.

The histograms revealed that the advertising budgets for different media channels are distributed unevenly across observations. The sales variable showed an approximately normal distribution with a moderate spread.

Purpose:

- Understand data distribution
- Identify skewness
- Detect unusual observations



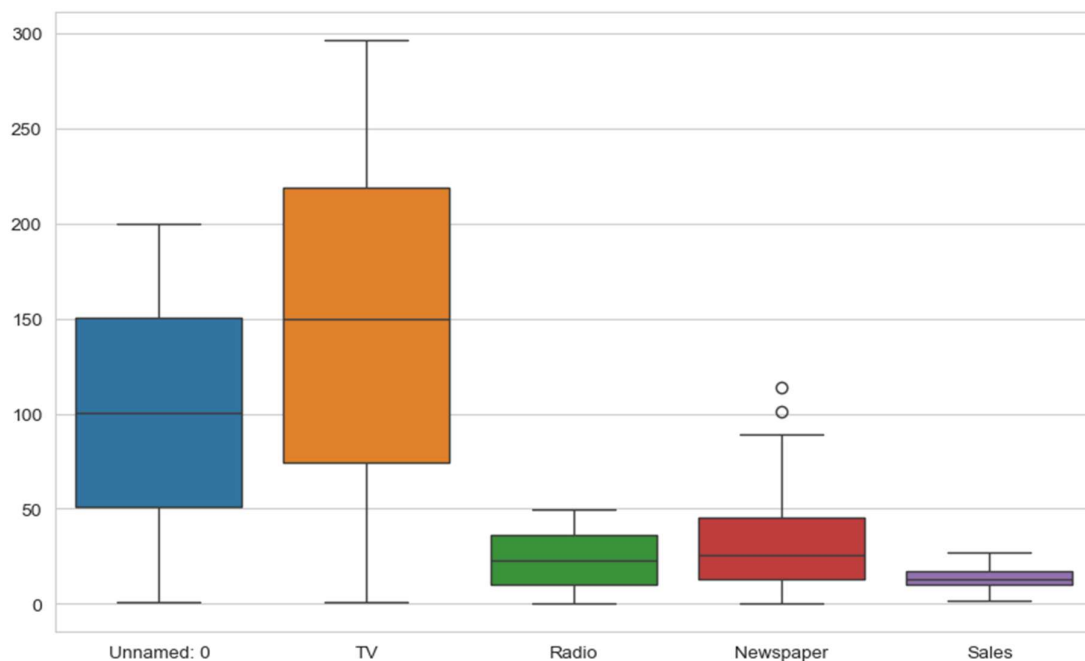
2. Box Plot

Box plots were created to identify the spread and variability of each feature and to detect potential outliers.

The box plots indicated the presence of a few outliers in the Newspaper advertising variable, while TV and Radio expenditures showed relatively consistent distributions.

Purpose:

- Detect outliers
- Understand data dispersion
- Compare variable ranges



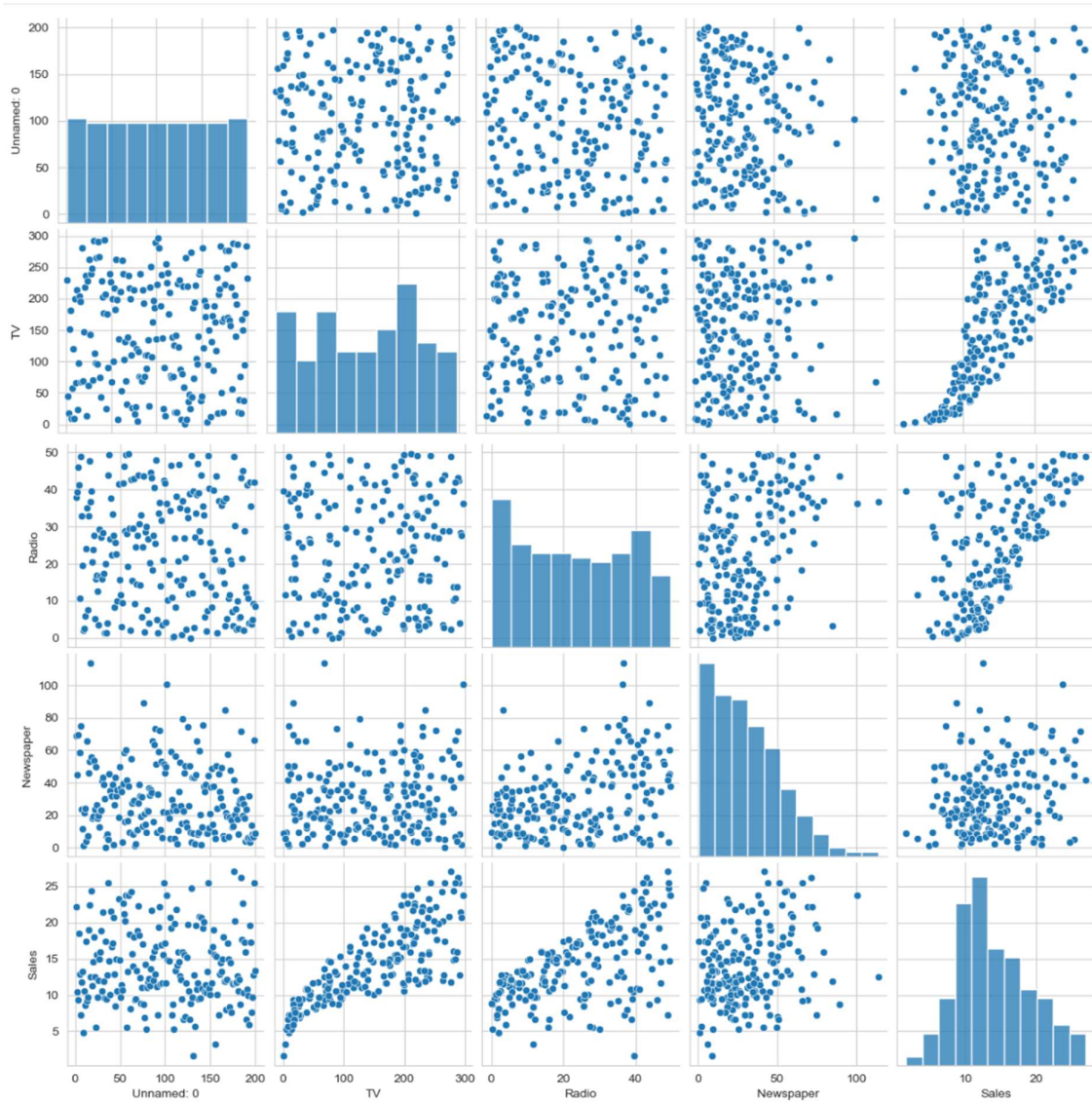
3. Pair Plot

Pair plots were generated to visualize pairwise relationships among all numerical variables.

The pair plot revealed strong positive relationships between TV advertising and Sales, as well as a moderate relationship between Radio advertising and Sales. Newspaper advertising showed a weak association with Sales.

Purpose:

- Analyze relationships between variables
- Observe data trends
- Identify clusters and patterns



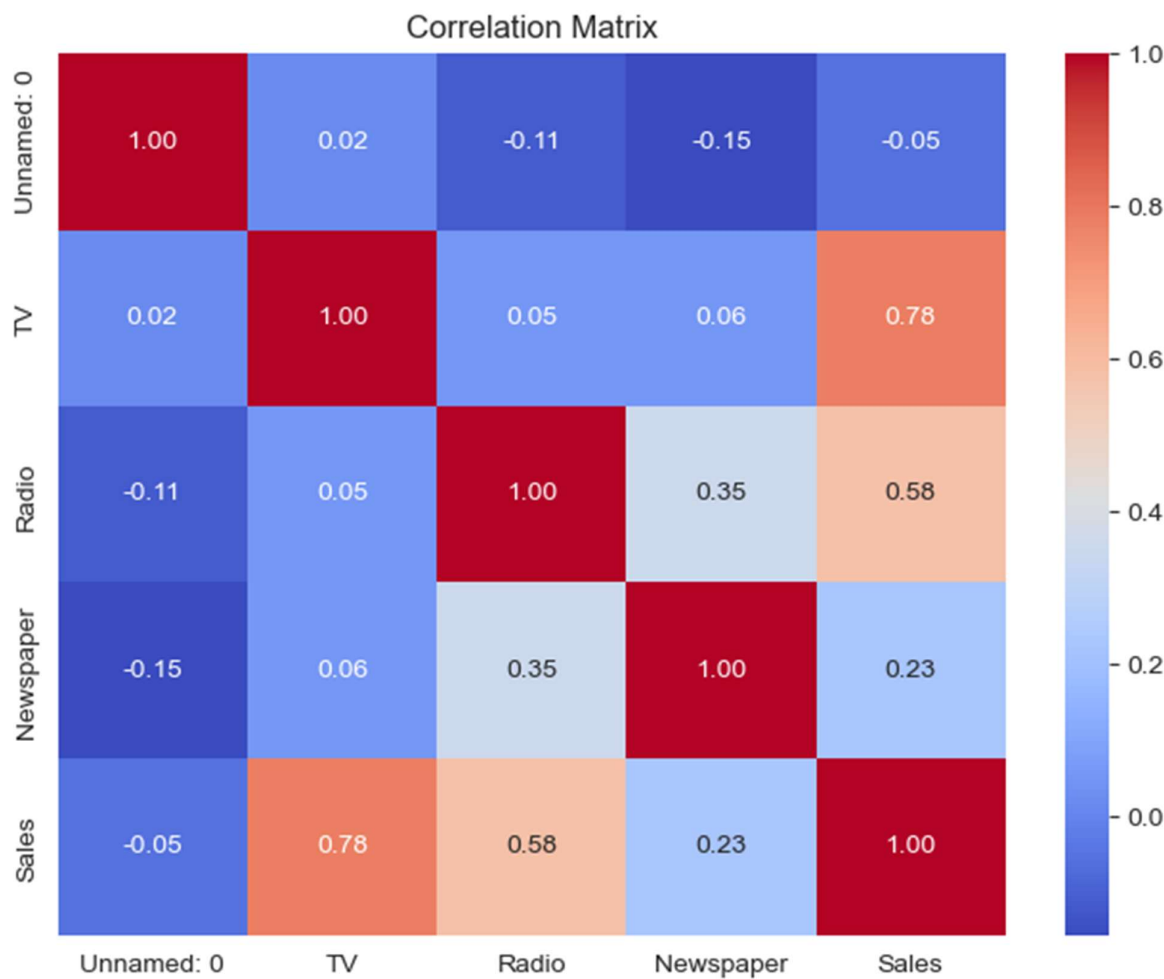
4. Correlation Heatmap

A correlation heatmap was created to quantify the strength of relationships among variables.

The heatmap showed that TV advertising has the highest positive correlation with Sales, followed by Radio advertising. Newspaper advertising exhibited a relatively weak correlation.

Purpose:

- Measure feature relationships
- Identify influential variables
- Support feature selection



5. Scatter Plot: TV vs Sales

The scatter plot between TV advertising expenditure and Sales demonstrated a strong positive linear relationship.

As the investment in TV advertising increased, sales also increased significantly.

6. Scatter Plot: Radio vs Sales

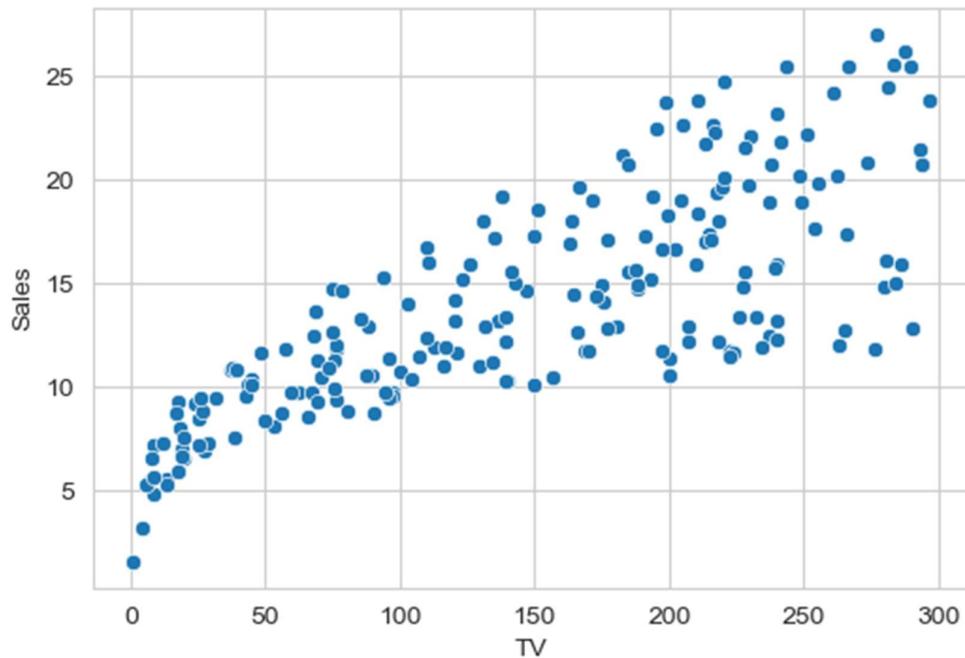
The scatter plot between Radio advertising expenditure and Sales showed a moderate positive relationship.

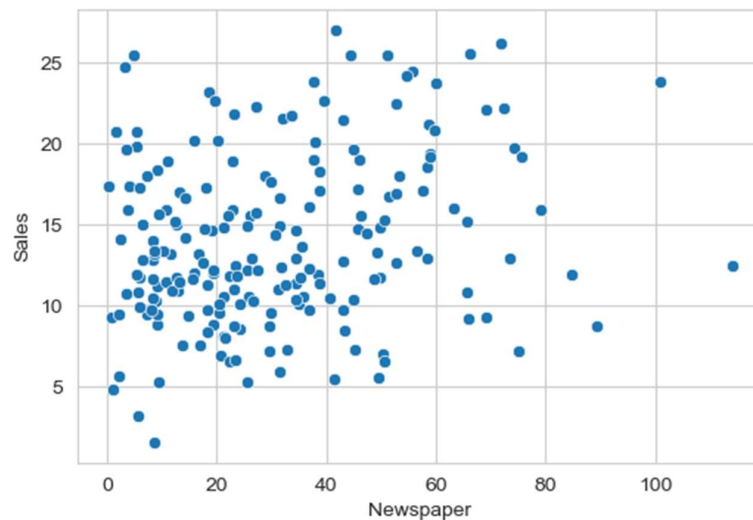
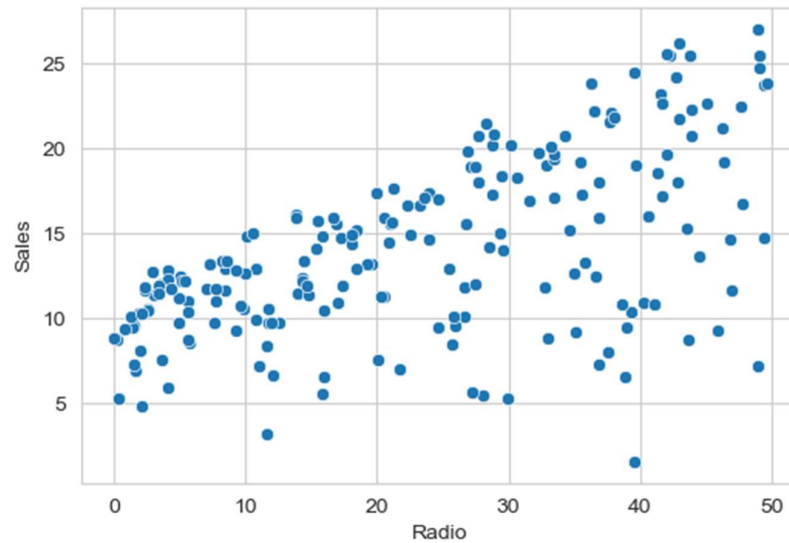
Higher spending on Radio advertisements generally resulted in increased sales, although the relationship was less strong compared to TV advertising.

7. Scatter Plot: Newspaper vs Sales

The scatter plot between Newspaper advertising expenditure and Sales indicated a weak relationship.

Changes in Newspaper advertising budgets did not significantly affect product sales.





Summary of Visual Findings

The visual analysis confirmed that TV and Radio advertising channels contribute more effectively to sales growth than Newspaper advertising. Among all features, TV advertising emerged as the most influential factor affecting sales performance.

The visualizations also helped verify data quality by identifying outliers and confirming the absence of major anomalies within the dataset.